8-5. Model comparison: Analysis of deviance and Akaike Information Criterion

- Recall the deviance (sometimes called the *residual deviance*) is defined as

$$D = -2\left(l_{\widehat{\boldsymbol{\beta}}} - l_{sat}\right)$$

  where
  - $l_{\widehat{\boldsymbol{\beta}}}$ is the maximised log-likelihood for the regression model of interest
  - l_{sat} is the maximised log-likelihood for the saturated (or maximal) model

- Note that we always have $l_{\widehat{\boldsymbol{\beta}}} \leq l_{sat}$, so that deviance values are non-negative.

- The closer the model of interest is to the full (saturated) model, the smaller the deviance will be, and the better the fit.
- **Good models have small deviance**
- How do we check if deviance is "small enough"? If model fits adequately, then

$$D \sim \chi^2_{n-p} \quad \text{(approximately)}$$

where n is the number of observations and p the number of fitted parameters.

Null deviance

- Consider now the *null model*, in which all mean response are assumed to be equal, so $\mu_i = \mu$ for all i.
- The deviance for this model is called the *null deviance*.
- Notice that in general,

$$\begin{aligned} & l_{null} \leq l_{\widehat{\boldsymbol{\beta}}} \leq l_{sat} \\ \Rightarrow \quad & -2\, l_{sat} \leq -2\, l_{\widehat{\boldsymbol{\beta}}} \leq -2\, l_{null} \\ \Rightarrow \quad & D_{null} \geq D_{\widehat{\boldsymbol{\beta}}} \geq 0 \end{aligned}$$

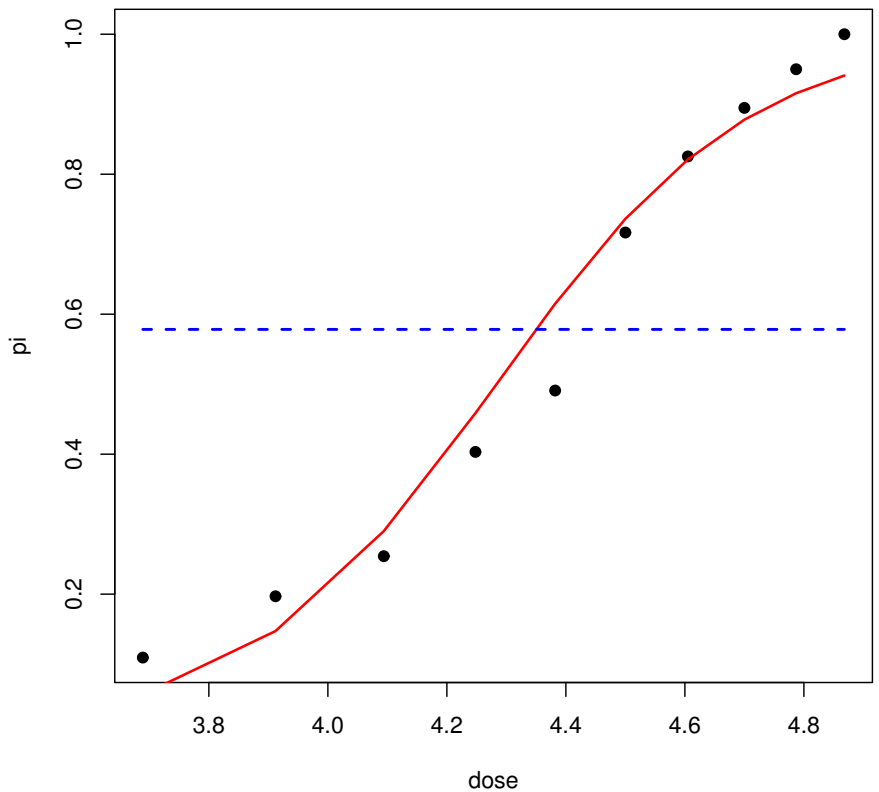
- If the response is **not related** to the covariate, the model (residual) deviance will not be much smaller than the null deviance.

Example 8.3

Recall the data of example 8.3: 10 groups of insects exposed to different doses of a drug. We have 3 *nested* models:

- The saturated (full) model
  Fitted values are $\hat{\pi}_i = d_i/n_i$
- The logistic model
  Fitted values obtained by using the logistic function with dose as a covariate.
- The null model
  Fitted values are equal to the common (mean) estimate
  $\hat{\pi} = \left(\sum_i d_i\right) / \left(\sum_i n_i\right)$

```
> pi.mean = sum(numdead)/sum(groupsize)
> plot(propdead~dose, ylab = "pi", pch=19)
> lines(dose, fitted(drugmodel), col="red", lwd=1.8)
> lines(dose, rep(pi.mean,length(dose)), col="blue", lwd=2.0, lty=2)
```



Plot shows all 3 models. Maximal model = black dots; logistic regression model = red curve; null model = blue dashed line.

Recalling that Binomial log-likelihood is

$$l\left(\boldsymbol{\pi};\boldsymbol{d}\right) = \sum_i \left(\log \binom{n_i}{d_i} + d_i \log\left(\pi_i\right) + \left(n_i - d_i\right) \log\left(1-\pi_i\right) \right)$$

we can compute

$$\begin{aligned} l_{sat} &= -18.033 \\ l_{\widehat{\boldsymbol{\beta}}} &= -26.65 \\ l_{null} &= -167.065 \end{aligned}$$

Deviances are thus

$$\begin{aligned} \text{(Residual) deviance} &= -2\left(l_{\widehat{\boldsymbol{\beta}}} - l_{sat}\right) \\ &= -2\left(-26.65 - (-18.033)\right) = 17.234 \\ \text{Null deviance} &= -2\left(l_{null} - l_{sat}\right) \\ &= -2\left(-167.065 - (-18.033)\right) = 298.07 \end{aligned}$$

Confirm deviance values from R output:

```
> drugmodel

Call:  glm(formula = propdead ~ dose, family = binomial, weights = groupsize)

Coefficients:
(Intercept)         dose
    -20.272        4.733

Degrees of Freedom: 9 Total (i.e. Null);  8 Residual
Null Deviance:      298.1
Residual Deviance: 17.23        AIC: 57.3
```

Analysis of deviance: model comparison

- Compare the fit of two *nested* models:
  M_1 with variables $x_1, \ldots, x_p$
  M_2 with variables $x_1, \ldots, x_p, x_{p+1}, \ldots, x_{p+k}$
- Test the significance of the additional k parameters in M_2:

$$\begin{aligned} H_0 &: \beta_{p+1} = \beta_{p+2} = \cdots = \beta_{p+k} = 0 \\ \text{vs. } H_1 &: \beta_{p+j} \neq 0 \text{ for at least one } j \in \{1, \ldots, k\} \end{aligned}$$

- We can test the significance of the additional k parameters in M_2 using the change in deviance. Under H_0,

$$D = D_1 - D_2 \sim \chi^2_k \qquad \text{(approximately)}$$

- Reject H_0 for large D (one-sided test)
- The 'bigger' model M_2 will always fit better than the 'smaller' nested model M_1. Not rejecting H_0 means that M_2 does not give *significantly* better fit than M_1, hence prefer M_1.

Example 8.3 (continued)

- Recall for example 8.3, for logistic regression model M_2 and null model M_1, we computed
  residual deviance $D_2 = 17.234$ on 8 degrees of freedom,
  null deviance $D_1 = 298.07$ on 9 degrees of freedom.
- Comparing the two models, change in deviance is
  $D_1 - D_2 = 298.07 - 17.234 = 280.836$ on $9 - 8 = 1$ d.f.
- Testing whether logistic regression model fits significantly better than the null model, p-value is
  `1 - pchisq(280.836, df=1) = 0` (to machine precision).
- Logistic regression model fits very significantly better than the null model. It seems that the covariate (dose of drug) does have an effect upon the response (insect mortality rate).

Although our linear logistic model fits better than the null model, we previously saw that it is still not a very good fit. Try quadratic model.

```
> drugmodel2 = glm(propdead~dose+I(dose^2), weights=groupsize, family=binomial)
> summary(drugmodel2)

Call:
glm(formula = propdead ~ dose + I(dose^2), family = binomial,
    weights = groupsize)

Deviance Residuals:
     Min        1Q    Median        3Q       Max
-0.79342  -0.33844  -0.00908   0.34745   1.68311

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   47.154     18.821   2.505 0.012232 *
dose         -26.989      8.913  -3.028 0.002461 **
I(dose^2)      3.713      1.051   3.531 0.000414 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

    Null deviance: 298.0666  on 9  degrees of freedom
Residual deviance:   4.5903  on 7  degrees of freedom
AIC: 46.656
```

Now consider models M_1 (linear) and M_2 (quadratic).

$$\begin{aligned} M_1 &: \eta = \beta_0 + \beta_1 x \\ M_2 &: \eta = \beta_0 + \beta_1 x + \beta_2 x^2 \end{aligned}$$

('Nested': start with M_2, impose the constraint $\beta_2 = 0$, obtain model M_1 nested within M_2.)

Deviance: $D_1 = 17.23$ on 8 df, and $D_2 = 4.590$ on 7 df

Change in deviance:

$$D_1 - D_2 \;=\; 17.23 - 4.59 \;=\; 12.64 \text{ on 1 d.f.}$$

Testing whether M_2 provides significantly better fit than M_1,

$$\text{p-value} \;=\; \Pr\left(\chi^2_1 > 12.64\right) \;=\; 0.0004$$

Addition of quadratic term provides significant improvement in the fit of the model.

```
> anova(drugmodel,drugmodel2,test="Chi")
Analysis of Deviance Table

Model 1: propdead ~ dose
Model 2: propdead ~ dose + I(dose^2)
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1         8    17.2340
2         7     4.5903  1   12.644 0.0003768 ***
```

See `?anova.glm`

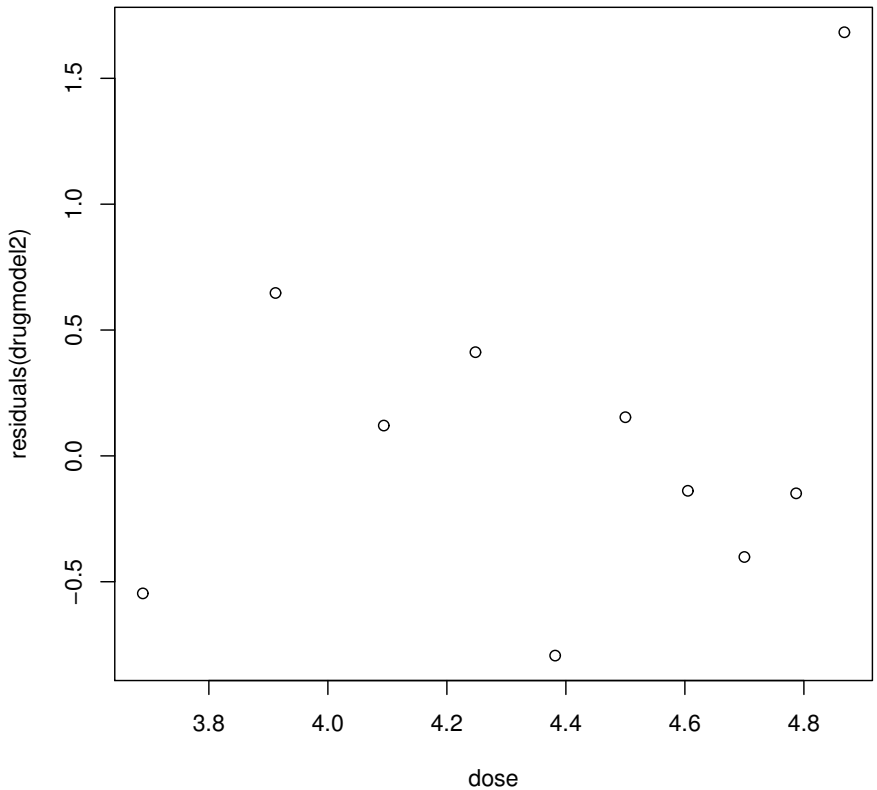
Using quadratic model M_2, for dose x we have

$$\begin{aligned} \hat{\eta} &= 47.15 - 26.99x + 3.71x^2 \\ \hat{\pi} &= \frac{\mathrm{e}^{\hat{\eta}}}{1 + \mathrm{e}^{\hat{\eta}}} \end{aligned}$$

For instance, for dose $x = 4.0$, fitted insect death probability is

```
> predict(drugmodel2,data.frame(dose=4),type="response")
0.1978144
```

Plot residuals versus dose for quadratic model.



The residual plot looks much more satisfactory than before. Plot suggests that this quadratic model provides a better description of our data.

Analysis of deviance

Note on use of deviance in `R`:

The deviance, as defined for GLMs here, includes the *scale* parameter ϕ

`R` provides in its output the **residual deviance**, where

$$D = \frac{\text{Residual Deviance}}{\phi}$$

When ϕ is unknown, use an estimate of it for computing D (e.g. the '`dispersion parameter`' in `R`.)

You can verify that:

- For Binomial(n, p), $\phi = 1$
- For Poisson(λ), $\phi = 1$
- For N(μ, σ^2), $\phi = \sigma^2$
- For Gamma$(a, a/\lambda)$, $\phi = 1/a$

Akaike Information Criterion (AIC)

- Another measure of fit is the **Akaike Information Criterion**:

$$AIC \quad = \quad -2\, l_{\widehat{\boldsymbol{\beta}}} + 2p$$

  where p is again number of parameters fitted.

- Use AIC to compare the fit of different models.
- **Good models should have small AIC**
- AIC also accounts for number of parameters in model; it penalises models with large number of parameters

Akaike Information Criterion

Consider two nested models M_1 and M_2, with M_2 having more parameters. Rule of thumb:

- If $AIC_{M_1} - AIC_{M_2} > 2$ conclude that M_2 provides substantial improvement as compared to M_1

Only useful for **comparing** models.

In example 8.3, for our linear logistic model M_1 we have

$$AIC \quad = \quad -2\,(-26.65) + 4 \quad = \quad 57.28$$

For the quadratic logistic model M_2, we have $AIC = 46.7$. Comparing the two models,

$$AIC_{M_1} - AIC_{M_2} \quad = \quad 57.3 - 46.7 \quad = \quad 10.6 \quad > \quad 2$$

suggesting M_2 should be preferred.

References

- Boland, P.J. (2007) *Statistical and Probabilistic Methods in Actuarial Science*, Chapman and Hall/CRC, Boca Raton.
- Dobson, A.J. (2002) *An Introduction to Generalized Linear Models*, 2nd edn, Chapman and Hall, London.
- Faraway, J. (2006) *Extending the Linear Model with R : Generalized Linear, Mixed Effects and Nonparametric Regression Models*, Chapman & Hall/CRC, Boca Raton.
- McCullagh, P. and Nelder, J.A. (1989) *Generalized Linear Models*, 2nd edn, Chapman and Hall, London.